

Virtual Sets

mark

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Definition 0.1 Given an injective map $f : X + A \rightarrow Y + A$, where A is finite, we define a new map

$$(f - A) : X \rightarrow Y$$

$$(f - A)(x) = \begin{cases} y & \text{if } f(x) = y : Y \\ f_A(a) & \text{if } f(x) = a : A \end{cases}$$

Where we define a helper for each $S \subseteq A$

$$f_S : S \rightarrow Y$$

$$f_S(a) = \begin{cases} y & \text{if } f(a) = y : Y \\ f_{S \setminus a}(a') & \text{if } f(a) = a' : A \end{cases}$$

Note by injectivity of f we have that a and a' are different, and since A , we will eventually hit an element of Y .

We can prove a number of addition rules for these functions

Lemma 0.2 Given $f : X \rightarrow Y$ we have $(f + A) - A = f$.

Proof. For any $x : X$ we have $(f + A)(\text{inl } x) = \text{inl } f(x) : Y$. Hence we see by definition that $((f + A) - A)(x) = f(x)$. $\square$

Lemma 0.3 Given $f : X + A \rightarrow Y + A$ with A finite, for any set B we have

$$(f - A) + B = (f + B) - A$$

Proof. Note for any $b : B$ we have $((f + B) - A)(b) = b$ by definition. and for $x : X$ we have $(f + B)(x) = f(x)$, and similarly for any $a : A$. Unwinding definitions makes it clear that f_A and $(f + B)_A$ are the same function, and further that $((f + B) - A)(x) = (f - A)(x)$. $\square$

Lemma 0.4 Given $f : X + A + B \rightarrow Y + A + B$ we have $(f - B) - A = f - (A + B)$.

Proof. Need to come up with a better way of seeing this but we can redefine the notion of subtraction in the following way: For an easier way to see this think of a function $X + A \rightarrow Y + A$ as a kind of graph on the vertex set $X + A + Y + B$, with edges $x \rightarrow f(x)$. Then we can take a graph quotient identifying the two copies of A and the left copy of A with its image under f . So $\text{inl } a \sim \text{inr } a$ and $\text{inl } a \sim f(a)$. Taking this graph quotient results in the graph of the function $f - A : X \rightarrow Y$. This is a kind of “small step” reduction process to produce $f - A$. Then we can see clearly that $f - B - A = f - (A + B)$ by just checking the two equivalence relations are the same, which they clearly are. $\square$

Lemma 0.5 We have $(g + A) \circ f - A = g \circ (f - A)$. Similarly $g \circ (f + A) = (g - A) \circ f$.

Given these we are now in a position to work with virtual sets.

Definition 0.6 The category of virtual sets is given by

- Objects pairs of finite sets (X^+, X^-)
- Morphisms $X \rightarrow Y$ given by an injective map $X^+ + Y^- \rightarrow Y^+ + X^-$.
- Identities given by identity maps
- Composition given by the following construction. Given $f : X^+ + Y^- \rightarrow Y^+ + X^-$ and $g : Y^+ + Z^- \rightarrow Z^+ + Y^-$ we define a map

$$\begin{aligned} h : X^+ + Z^- + Y^- &\xrightarrow{f+Z^-} Y^+ + Z^- + X^- \\ &\xrightarrow{g+X^-} Z^+ + Y^- + X^- \\ &\xrightarrow{\sim} Z^+ + X^- + Y^- \end{aligned}$$

We will continue to abuse coherences and just write h as $(g + X^-) \circ (f + Z^-)$. The composition is given by $g \circ f := h - Y^-$.

We now check that composition is associative.

Proposition 0.7 Composition is associative.

Proof. Let $f : A \rightarrow B$, $g : B \rightarrow C$ and $h : C \rightarrow D$. Then we calculate using the lemmas before

$$g \circ f = (g + A^-) \circ (f + C^-) - B^-$$

$$\begin{aligned} h \circ (g \circ f) &= (h + A^-) \circ (g \circ f + D^-) - C^- \\ &= (h + A^-) \circ (((g + A^-) \circ (f + C^-) - B^-) + D^-) - C^- \\ &= (h + A^-) \circ (((g + A^-) \circ (f + C^-) + D^-) - B^-) - C^- \\ &= (h + A^-) \circ ((g + A^- + D^-) \circ (f + C^- + D^-) - B^-) - C^- \\ &= (h + A^- + B^-) \circ ((g + A^- + D^-) \circ (f + C^- + D^-)) - B^- - C^- \\ &= (h + A^- + B^-) \circ (g + A^- + D^-) \circ (f + C^- + D^-) - (C^- + B^-) \end{aligned}$$

Which is precisely the natural “triple composition” given between three morphisms.

Similarly we have

$$h \circ g = (h + B^-) \circ (g + D^-) - C^-$$

$$\begin{aligned} (h \circ g) \circ f &= (h \circ g + A^-) \circ (f + D^-) - B^- \\ &= (((h + B^-) \circ (g + D^-) - C^-) + A^-) \circ (f + D^-) - B^- \\ &= (((h + B^-) \circ (g + D^-) + A^-) - C^-) \circ (f + D^-) - B^- \\ &= ((h + B^- + A^-) \circ (g + D^- + A^-) - C^-) \circ (f + D^-) - B^- \\ &= ((h + B^- + A^-) \circ (g + D^- + A^-)) \circ (f + D^- + C^-) - C^- - B^- \\ &= (h + B^- + A^-) \circ (g + D^- + A^-) \circ (f + D^- + C^-) - (B^- + C^-) \\ &= (h + B^- + A^-) \circ (g + D^- + A^-) \circ (f + D^- + C^-) - (C^- + B^-) \end{aligned} \quad \square$$

Remark 0.8 Here is a relatively convincing reason why this category might not be correct: two objects are only isomorphic if their pairs of objects are isomorphic. This is unlike the intuition that we would expect $(3, 2) \simeq (2, 1)$.

We prove this now. Suppose $X \simeq Y$ in the category. Suppose they don't have the same size (this is fine because of decidability). WLoG suppose $|Y^+| < |X^+|$. Call $f : X \rightarrow Y$ and $g : Y \rightarrow X$ the functions in the isomorphism. Then note that for some $x^+ : X^+$ we must have $f(x^+) \in X^-$, due to size. Then by definition $(g \circ f)(x^+) = f(x^+)$, since g acts constantly on this copy of X^- . Then clearly $(g \circ f)(x^+) = f(x^+) \neq x^+$. So we must have $|Y^+| \geq |X^+|$. Arguing dually we see that $|Y^+| = |X^+|$ and so we deduce $|Y^-| = |X^-|$.